

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^3, T(x, y) = (x-5, 2x+y, 0)$$

Transformar em vetor

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x-5 \\ 2x+y \\ 0 \end{bmatrix}$$

Testar com regra de aditividade

$$T(\underline{\vec{u}} + \underline{\vec{v}}) = T(\underline{\vec{u}}) + T(\underline{\vec{v}})$$

Escolher qualquer vetor

$$\underline{\vec{u}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \underline{\vec{v}} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$T(\underline{\vec{u}} + \underline{\vec{v}}) = T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = T\left(\begin{bmatrix} 3 \\ 5 \end{bmatrix}\right) = \begin{pmatrix} 3-5 \\ 2 \times 3 - 5 \\ 0 \end{pmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$T(\underline{\vec{u}}) + T(\underline{\vec{v}}) = T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) + T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) =$$

$$= \begin{bmatrix} 2-5 \\ 2 \times 2 - 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 1-5 \\ 2 \times 1 - 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -7 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \quad \checkmark$$